

proposal from a solver
loop + pins {book root, catalogue root}

pins equal ours?

→ refuse

re-derive every leg
against the *current* book

→ refuse

chaining, product > 1 ,
nothing already filled

→ refuse

commit all fills + loop
under one root

the solver is a hint, never an authority. Atomicity is one recordstore commit: either every leg fills or none does.